Conversion of Camphene to Camphor (Day 3)

Brooke Arnold

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Introduction

In this experiment, the borneol product yielded from last week's experiment was mixed with glacial acetic acid, acetone, and aqueous bleach and placed in a water bath in order extract camphor. The solution was then neutralized, washed, dried over a drying agent, and set over a water bath in order to yield camphor crystals.

Materials and Methods

About 0.3 g of last week's borneol product and 0.3 mL of glacial acetic acid were combined in a 10 mL long-neck round-bottom flask. A reflux with a distillation column was assembled, then the apparatus was placed in a hot water bath with a temperature between 40-50 °C. Once the reagents dissolved, 6 mL of aqueous bleach was added dropwise through the top opening of the condenser on the apparatus over the course of 35 minutes. The apparatus was agitated after each addition of bleach so proper mixing could occur. The reaction was allowed to continue for 15 minutes after the addition of the last drop of bleach. The mixture was allowed to cool to room temperature and the camphor was extracted with two 2 mL portions of DCM. A centrifuge tube was used in order to make the extraction more efficient. The combined methylene chloride layers were neutralized with ~2 mL of saturated sodium bicarbonate, then the organic solution was washed with first 2 mL of 5% sodium bisulfite and then 2 mL of water. The organic solution was dried over anhydrous sodium sulfate, then the remainder of the DCM was evaporated over a hot water bath with a temperature not exceeding 50°C. The collected crystals were then weighed.

Results

Weight of Dry Watch Glass (g)	43.021
Weight of Watch Glass and Crystals (g)	44.275
Weight of Crystals (g)	0.196

Table 1. Weight of watch glass, watch glass and crystals, and crystals.

Calculations

$$\frac{0.3\ g\ borneol}{1} x \frac{1\ mol\ borneol}{154.25\ g\ borneol} = 0.001945\ mol\ borneol$$

$$\frac{0.001945\ mol\ camphor}{1} x \frac{152.23\ g\ camphor}{1\ mol\ camphor} = 0.296\ g\ camphor$$

$$\frac{0.254\ g\ camphor}{0.296\ g\ camphor} x\ 100\% = 85.81\%\ yield$$

Discussion

In this experiment, 0.196 g of camphor crystals were isolated from a reaction between borneol, glacial acetic acid, and other compounds using a hot water bath in order to encourage the reaction. Based on the appearance and scent of the crystals, coupled with the crystals not evaporating away while the excess DCM was heated from the sample, it can be concluded that the identity of the crystals produced were camphor. The identity of the crystals could be further confirmed by taking the boiling point of the crystals. Possible sources of error in this experiment include insufficient drying of the sample with anhydrous sodium sulfate, or leaving some organic sample left behind during the extraction phase. Of the mentioned errors, it was most likely that some organic sample was left inside the centrifuge tube during the extraction phase, since it was hard to differentiate between each layer as the pipette approached the borderline. This would help to explain the percent yield below 100%. One way to resolve this error would be to use a

method of washing the organic sample that is more precise, or by somehow adding color to one of the layers in order to make distinguishing them easier.

References

Keller L, Joseph K, and Miguel S. 2023 Fall. CHEM 2081 Organic Chemistry 2 Lab Manual [lab manual]. Auburn (Al): Auburn University.